Three cases: four reductions in the cut analysis and the final count

A third research note on "The maximum number of cycles in a graph of large rank"

Fable 5.2, for KNT — Research note of 12 November 2026

This note answers the request of 12 November 2026 to search for more efficiencies. It has four reductions of the proof in [M], outside the last section, and a list of attempts that failed. The numbers of the statements of [M] are those of the draft of 42 pages. [N1] and [N2] are the first two notes, `one-scale.pdf` and `girth-five.pdf`. The manuscript is not changed.

The result in eight sentences

1. *The gap theorem is a corollary.* A bridgeless 2-pole is a capped ladder if and only if the graph J outside a shortest path between its ports is a forest; this is what the proof of [M, Lemma 5.8] shows. If J has a cycle and the pole has many paths, then J has a short cycle, by Step 2 of the proof of [M, Proposition 5.3], and [M, Lemma 5.1] gives a gap. So [M, Lemma 5.9] (a third port, four cases) and the induction in [M, Theorem 5.10] are not necessary (Theorem 2.3).
2. *A cycle of length four in the place of a cut of size four.* By [N2], the last case needs girth five and no hypothesis on cuts. So [M, Theorem 6.8] is necessary only for a graph with a cycle C of length four. Then one side of the cut is C, the count is explicit, and it is the sum of the counts for two caps (equation (3.4)). [M, Lemma 6.4] (gluing) and all steps for the second side are not necessary (Theorem 3.3).
3. *Three cases.* A loop gives a cyclic cut of size one, so the case of a loop is a part of [M, Proposition 6.3]. The proof of the main theorem then has three cases: a cyclic cut of size at most three; a cycle of length four; girth at least five.
4. *One count.* The count with a marked set of eight vertices, which the last case of [M] uses, also gives $\psi(r) \le (\frac{1}{2} + o(1))2^r$ with a marked set of $4 \log r$ vertices. So the edge recursion in the proof of [M, Corollary 3.7] is not necessary (Lemma 5.1, Corollary 5.2).
5. *Measured.* A copy of the paper with the lean form of [N2] and the four reductions builds: 41 pages and 1587 typeset lines (now 42 and 1624), 39 numbered statements (now 40), and three cases (now five). Without the first reduction the copy has 42 pages and 1607 lines.
6. *The price.* The gap constant is $\gamma_0 = 2^{-20\Theta}$ with a very large Θ, in the place of $\frac{1}{16}$. So Section 6 has a small number $\varepsilon_0 = \gamma_0/20$ in the place of $\frac{1}{1000}$, and the gap theorem now depends on the forest estimate. The text differs more from the formal proof.
7. *Tests.* A script tests the ladder criterion, the count for a cycle of length four, the count with a marked set, and the arithmetic of the rigidity lemma on small graphs (Section 8). All tests

pass.

8. *What failed.* I found no way to remove [M, Proposition 6.3], the lemma on a full star, or the base of the lemma on pieces, and no polynomial bound for pieces at girth five without an induction (Section 7).

> **NOTATION | the status of this note**
>
> - *Established on paper.* Sections 2 to 5 contain complete proofs. They use statements of [M] with their numbers, and Section 3 uses [N2, Theorem 5.1] for the order of the cases. I checked each step against these statements.
> - *Tested by computer.* Section 8. The tests cover small graphs and the arithmetic. They do not replace a proof.
> - *Not done.* There is no formal proof. No other person or model has read this note.
> - *The manuscript.* It is unchanged. The script `apply_fewer_cases.py` applies the four reductions to a copy that `apply_girth_five.py` has changed.

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1 What changes

Part of [M]	Now	With this note
[M, Lemma 5.9] (a third port)	a lemma with four cases	not necessary
[M, Theorem 5.10] (gap)	induction on the rank, constant $\frac{1}{16}$	eight lines from [M, Proposition 5.3] and [M, Lemma 5.8], constant γ_0
[M, Lemma 5.1], last statement	a bound with the number of components of $N - E(Z)$	the bound $2^b - 2^{b-\lvert Z\rvert}$
[M, Lemma 6.4] (gluing)	a count for two 4-poles	not necessary; a count for a cycle of length four
[M, Theorem 6.8]	a cyclic cut of size four and no smaller one; two sides	a cycle of length four and no cyclic cut of size at most three; one side
Case 1 of [M, Section 9] (a loop)	a case	the case $d = 1$ of [M, Proposition 6.3]
[M, Corollary 3.7] (leading term)	a recursion over the edges of a shortest cycle	the count with a marked set
The count in the last case	in the proof of the main theorem	Lemma 5.1, used two times

The reductions 1, 3 and 4 do not depend on [N2]. Reduction 2 needs the last case in the form of [N2]: a graph with a cyclic cut of size four and girth at least five then belongs to the last case. The four reductions are independent of each other, with one exception: the constants of Section 3 are those of the new gap theorem.

2 The gap theorem without induction

2.1 The statements of [M] that we use

A *pole* is a connected multigraph without loops in which every vertex has degree two or three; its vertices of degree two are its *ports*. For two ports s, t of a pole N of rank b, $P_{st}(N)$ is the number of paths from s to t, and the pair is *full* if $P_{st}(N) = 2^b$. We use these statements.

- [M, Lemma 5.1]. Every $\{s,t\}$-join is the union of a path from s to t and of cycles, pairwise vertex-disjoint. So the pair is full if and only if every cycle meets every path from s to t.
- [M, Lemma 5.2]. If $P_{st}(N) > 2^{b-1}$, replace the other ports by edges. The result consists of bridges and of bridgeless 2-poles $N_1, \dots, N_q$ of ranks $b_i \geq 1$ with $\sum_i b_i = b$, $P_{st}(N) = \prod_i P(N_i)$ and $P(N_i)\,2^{-b_i} \geq P_{st}(N)\,2^{-b}$.
- The notation of [M, Section 5.2] for a bridgeless 2-pole N of rank $b \geq 2$: the layers L_j (the vertices at distance j from s), the distance ℓ from s to t, a shortest path $A = w_0 \cdots w_\ell$, and $J = N - V(A)$. The pole has $2b$ vertices and no cut vertex, and every vertex of A has at most one edge that is not an edge of A.
- The count in Step 4 of the proof of [M, Proposition 5.3]. It uses no hypothesis on the number of paths: $\ell \leq b$; if J is not empty, then $\beta(J) \geq b - \ell$; if J is empty, then $2b = \ell + 1$.
- Steps 1 and 2 of the proof of [M, Proposition 5.3]. Let $P_{st}(N) \geq (\frac{1}{2} + \varepsilon)2^b$ and $\Theta = \Theta_{\varepsilon/2}(1)$. Then every layer has fewer than Θ vertices of J (Step 1, from [M, Lemma 4.1]). If the part of J in an interval of layers contains a cycle, choose a cycle Z there that meets as few layers as possible; then the layers that Z meets contain at most 20Θ vertices of J (Step 2).

The second item of Step 2, for the interval of all layers, is the following statement. In a copy of the paper it is a new item (iv) of [M, Proposition 5.3].

Lemma 2.1 (A short cycle). Let N be a bridgeless 2-pole of rank $b \geq 2$ with ports s, t and $P_{st}(N) \geq (\frac{1}{2} + \varepsilon)2^b$, where $0 < \varepsilon \leq \frac{1}{2}$. If J contains a cycle, then J contains a cycle with at most $20\,\Theta_{\varepsilon/2}(1)$ vertices.

2.2 Ladders

Lemma 2.2 (Ladders). Let N be a bridgeless 2-pole with ports s, t and rank b. If $b = 1$, then N consists of two vertices and two parallel edges, and the pair s, t is full. If $b \geq 2$, the following statements are equivalent:

(i) the pair s, t is full;

(ii) the graph $J = N - V(A)$ is a forest;

(iii) N is a capped ladder whose caps are s and t.

Proof. If $b = 1$, then N has two vertices, and it is a cycle. Let $b \geq 2$.

From (i) to (ii). By [M, Lemma 5.1], every cycle meets the path A.

From (ii) to (iii). This is the proof of [M, Lemma 5.8], which uses the full pair only to see that J is a forest. We repeat it. The count gives $\ell \leq b$. The graph J is not empty, because $2b > b + 1 \geq \ell + 1$. So $b - \ell \leq \beta(J) = 0$, and $\ell = b$. Then equality holds in every estimate of the count: every layer L_j with $0 < j < b$ consists of w_j and of one vertex y_j of J, there are no other vertices outside A, every vertex of A has a further edge that leads to J, and J is a tree. An edge joins vertices of equal or consecutive layers. So J is the path $y_1 \cdots y_{b-1}$, the further edge of s leads to y_1, that of t leads to y_{b-1}, and the further edge of w_j leads to a vertex $y_{\pi(j)}$ with $|\pi(j) - j| \leq 1$. Every y_i has degree three, so π is a permutation of $\{1, \dots, b-1\}$ that moves every number by at most one. A fixed point j is a single rung $w_j y_j$. An exchange of j and $j+1$ is a square with the left pair w_j, y_j and the right pair w_{j+1}, y_{j+1}. The other edges of A and J are the interfaces and the edges at the caps.

From (iii) to (i). Let the tiles have total weight W, so $b = W + 1$. A path from s to t passes the tiles in order. It enters a tile at a vertex of the left pair and leaves it at a vertex of the right pair. In a square there are two paths to each vertex of the right pair. In a single rung the path stays or crosses. So there are $2 \cdot 2^W = 2^b$ paths. ■

The count of the last step is in [M], before [M, Lemma 5.9].

2.3 The gap

Theorem 2.3 (Gap). There is a constant $\gamma_0 \in (0, \frac{1}{4}]$ with the following property. Let N be a pole of rank b, and let s, t be two of its ports. Then

$$P_{st}(N) = 2^b \qquad \text{or} \qquad P_{st}(N) \leq (1 - \gamma_0)\, 2^b.$$

Proof. Put $\Theta = \Theta_{1/8}(1)$ and $\gamma_0 = 2^{-20\Theta}$. Let $P_{st}(N) > (1 - \gamma_0)2^b$. By [M, Lemma 5.2], the path counts of the bridgeless 2-poles $N_1, \dots, N_q$ multiply, and $P(N_i) > (1 - \gamma_0)2^{b_i}$ for every i. So it suffices to show that the pair of ports of each N_i is full.

Let N be one of them, with rank b. If $b = 1$, the pair is full. Let $b \geq 2$. Suppose that J

contains a cycle. We have $P_{st}(N) \geq \frac{3}{4}\,2^b$. So J contains a cycle Z with at most 20Θ vertices, by Lemma 2.1 with $\varepsilon = \frac{1}{4}$. The cycle Z is disjoint from the path A. So A is an $\{s,t\}$-join of $N - E(Z)$, and $N - E(Z)$ has $2^{\beta(N-E(Z))} \geq 2^{b-|Z|}$ such joins. The union of each of them with $E(Z)$ is an $\{s,t\}$-join of N that is not a path. Hence $P_{st}(N) \leq 2^b - 2^{b-|Z|} \leq (1-\gamma_0)2^b$. This is a contradiction. So J is a forest, and the pair is full by Lemma 2.2. ■

IDEA | what the new proof uses, and what it does not use

The proof in [M] is an induction on the rank. It deletes the port s and needs a lemma on a 3-pole with one full pair ([M, Lemma 5.9]), which examines four positions of the third port in a capped ladder. That proof is elementary, and its constant is $\frac{1}{16}$. The new proof has no induction and no third port. It uses the layers of [M, Proposition 5.3], hence [M, Lemma 4.1] and the forest estimate. Its constant is not explicit in a useful sense: $\Theta = \Theta_{1/8}(1) \geq 2^{102}$.

The last statement of [M, Lemma 5.1] has the bound $2^b - 2^{b-|Z|+k-1}$, where k is the number of components of $N - E(Z)$. Only [M, Lemma 5.9] uses the number k. The proof above uses the bound with $k = 1$.

2.4 The constants of Section 6

In [M] the gap is $\frac{1}{16}$, and Section 6 has the numbers 0.501, 1.002, 2.004, 4.001 and 3.99, which come from $\varepsilon_0 = \frac{1}{1000}$. With the new theorem we put

$$\varepsilon_0 = \tfrac{1}{20}\,\gamma_0, \qquad K_0 = K_{\varepsilon_0}, \qquad \kappa(h) = 2K_0\,2^{-h/2},$$

with K_ε as in [M, Theorem 5.4]. Then $\varepsilon_0 \leq \frac{1}{80}$. [M, Lemma 5.7] holds with $\frac{1}{2}+\varepsilon_0$, $1+2\varepsilon_0$, $2+4\varepsilon_0$ and $1+2\varepsilon_0$ in the place of 0.501, 1.002, 2.004 and 1.002; its proof is the same. In the proof of [M, Proposition 6.3] the same replacements are sufficient: Case A needs $\frac{9}{32} + \frac{1}{2} + \varepsilon_0 < 1$ and $\frac{9}{32} + \frac{3}{2} + 3\varepsilon_0 < 2$, and Case B needs $\frac{1}{2} + 2\varepsilon_0 < 1$, $\frac{3}{2} + 4\varepsilon_0 < 2$ and $1 + 4\varepsilon_0 < 2$. The order of Section 5 changes: the constants and [M, Lemma 5.7] come after the gap theorem.

3 A cycle of length four in the place of a cut of size four

3.1 Why a cycle of length four is sufficient

In [M] the cases are ordered by cuts: a cyclic cut of size at most three; a cyclic cut of size four and no smaller one ([M, Theorem 6.8]); no cyclic cut of size at most four. The last case needs the hypothesis on cuts only to see that the girth is large. By [N2, Theorem 5.1], [M, Theorem 8.1] holds when $G - P$ has girth at least five, with no hypothesis on cuts, and the final count needs only that G has no bridge. So every cubic graph of large rank without a bridge and of girth at least five has fewer than 2^{r-1} cycles, also if it has a cyclic cut of size four. A graph without a cyclic cut of size at most three has no cycle of length at most three, for large rank. So the middle case is necessary only for a graph with a cycle of length four.

3.2 States, the cap, and rigidity

Let Y be a 4-pole of rank h with ports $1,2,3,4$. As in [M, Section 6.3], a *pairing* is one of the partitions $12|34$, $13|24$, $14|23$, and

$$a_{ij} = 2^{-h}P_{ij}(Y), \qquad u_\alpha = 2^{-h}f_{ij|kl}(Y)\ \ (\alpha = ij|kl), \qquad S = \sum_{ij} a_{ij}, \qquad F = \sum_\alpha u_\alpha,$$

where $f_{ij|kl}(Y)$ is the number of pairs of disjoint paths, one from i to j and one from k to l. Then $a_{ij} \le 1$ and $F \le 1$. With the constants of Section 2.4 and $\kappa = \kappa(h)$, [M, Lemma 5.7] reads

$$a_{ik} + a_{jk} \ge 1 + 2\varepsilon_0 \implies a_{ij} \le \kappa; \tag{3.1}$$

$$a_{ik} + a_{il} + a_{jk} + a_{jl} \ge 2 + 4\varepsilon_0 \implies u_{ik|jl} + u_{il|jk} \le \kappa; \tag{3.2}$$

$$a_{kl} > \kappa \implies a_{ij} + F - u_{ij|kl} < 1 + 2\varepsilon_0. \tag{3.3}$$

Fix h_0 such that $\kappa(h) \le \varepsilon_0$ for $h \ge h_0$.

Lemma 3.1 (Cap; [M, Lemma 6.5]). There is $h_1 \ge h_0$ with the following property. Let Y be a 4-pole of rank $h \ge h_1$, and put $z = 2^{-h}c(Y)$. Then $z + S + F - u_\alpha \le 4 + \varepsilon_0$ for every pairing α.

Proof. This is the proof of [M]. Let $\alpha = ij|kl$. Add two new vertices x and y and an edge between them, join x to the ports i, j and y to the ports k, l. The new multigraph $\hat{Y}$ is cubic of rank $h+3$, and it has at least $c(Y) + (S + F - u_\alpha)2^h$ cycles: the cycles of Y; a path of Y between two ports with the path of the new tree between its ends; and a pair of disjoint paths of Y whose pairing is not α with the four new edges at the ports. By [M, Corollary 3.7], $c(\hat{Y}) \le (\frac{1}{2} + \frac{1}{8}\varepsilon_0)2^{h+3} = (4 + \varepsilon_0)2^h$ for large h. ∎

Lemma 3.2 (Rigidity; [M, Lemma 6.6] with the new constants). Let Y be a 4-pole of rank $h \ge h_0$ with $S + F \ge 4 - 2\varepsilon_0$. Then three pairs with a common port are full, or the four pairs between two disjoint pairs of ports are full.

Proof. The proof is that of [M], with other numbers. Write η for ε_0; then $\kappa \le \eta \le \frac{1}{80}$. For a partition of the ports into two pairs, the *cross sum* is the sum of a_{ij} over the four pairs ij that are not pairs of the partition. Each pair occurs in two of the three cross sums. By Theorem 2.3, a pair ij with $a_{ij} > 1 - 20\eta$ is full.

Case 1: some cross sum is at least $2+4\eta$. Let it be the cross sum X of $12|34$. Then $a_{13}+a_{23} \ge 1+2\eta$ or $a_{14}+a_{24} \ge 1+2\eta$, so $a_{12} \le \kappa$ by (3.1). In the same way $a_{34} \le \kappa$. By (3.2), $u_{13|24}+u_{14|23} \le \kappa$. Put $u = u_{12|34}$. Then $S + F \le X + u + 3\kappa$. At most one of the four numbers in X is at most κ, because $X \ge 2 + 4\eta$. Let ij be one of the four pairs and kl the opposite pair. If $a_{kl} > \kappa$, then (3.3) gives $u < 1 + 2\eta - a_{ij}$.

Suppose that all four numbers are larger than κ. Then $u < 1 + 2\eta - m$, where m is the largest of them. So $S + F$ is less than the sum of the three smallest of them plus $1 + 5\eta$. As

$S + F \geq 4 - 2\eta$, each of the three is larger than $1 - 7\eta$. So all four pairs are full.

Suppose that one of the four numbers is at most κ, say a_{24}. Then $u < 1 + 2\eta - \max\{a_{14}, a_{23}\}$, and $S + F < a_{13} + \min\{a_{14}, a_{23}\} + 1 + 2\eta + 4\kappa \leq 3 + 6\eta$. This contradicts $S + F \geq 4 - 2\eta$.

Case 2: all cross sums are less than $2 + 4\eta$. Their sum is $2S$, so $S < 3 + 6\eta$. Hence $F > 1 - 8\eta$ and $S \geq 3 - 2\eta$, and the sum of two opposite numbers is more than $1 - 6\eta$.

Suppose that all six numbers are larger than κ. For a pairing $\alpha = ij|kl$, (3.3) gives $\max\{a_{ij}, a_{kl}\} + F - u_\alpha < 1 + 2\eta$, and the maximum is at least $\frac{1}{2} - 3\eta$. So $F - u_\alpha < \frac{1}{2} + 5\eta$. Adding this for the three pairings gives $2F < \frac{3}{2} + 15\eta$. This contradicts $F > 1 - 8\eta$, because $31\eta < \frac{1}{2}$.

So one number is at most κ, say a_{34}. Then $a_{12} > 1 - 7\eta$, so 12 is full. By (3.1), $a_{13} < 2\eta$ or $a_{23} \leq \kappa$; also $a_{14} < 2\eta$ or $a_{24} \leq \kappa$. If a number is less than 2η, its opposite is more than $1 - 8\eta$, hence full. If $a_{13} < 2\eta$, then 24 is full, so $a_{14} < 2\eta$, and 23 is full. Then the pairs $12, 23, 24$ are full. If $a_{23} < 2\eta$, then 14 is full, so $a_{24} < 2\eta$, and 13 is full. Then the pairs $12, 13, 14$ are full. ■

We also use [M, Lemma 6.7] (full star) without change: if the pairs $12, 13, 14$ are full in a 4-pole Y, then $u_\alpha(Y) \leq \frac{1}{2}$ for every pairing α. And we use [M, Corollary 5.11]: a 4-pole without bridge in which the four pairs between two disjoint pairs of ports are full is a chain of tiles between these two pairs.

3.3 The theorem

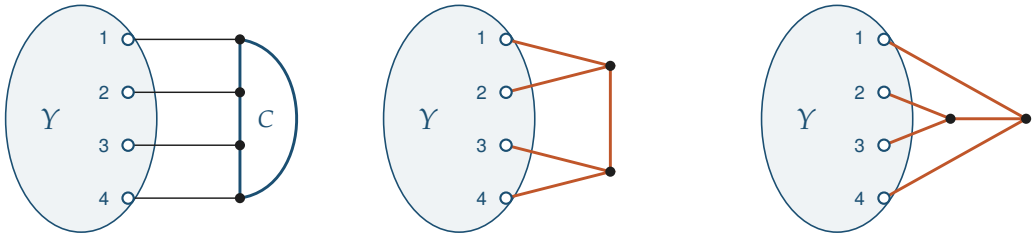


Figure 1. A cycle C of length four and the rest Y of the graph, and the two caps of Lemma 3.1 that take the place of C. By (3.4), the cycles of G that meet C and Y are as many as the cycles through the new vertices in the two capped graphs together.

Theorem 3.3. There is a constant r_4 with the following property. Let G be a connected cubic multigraph of rank $r \geq r_4$ that has no cyclic cut of size at most three and has a cycle of length four. Then G is a ring, or $c(G) < 2^{r-1}$.

Proof. Let $C = c_1c_2c_3c_4$ be a cycle of length four, and put $Y = G - V(C)$. By the remark at the start of [M, Section 6], every cut of size at most two is cyclic, and a cut of size three is cyclic unless one side is a single vertex. So G has no loop and no cut of size at most two, and every cut of size three has a single vertex on one side. The graph has $2r - 2$ vertices, so

a set of at most five vertices and its complement both have more than one vertex.

The rest is a 4-pole without bridge. No edge outside C joins two vertices of C: such an edge and a path of C form a cycle with at most three vertices, and the cut around these vertices has size at most three. So exactly four edges leave $V(C)$, one at each vertex. Let c_i be joined to the vertex i of Y. The vertices $1, 2, 3, 4$ are distinct: if two of them are one vertex v, the cut around $V(C) \cup \{v\}$ has size at most three. The multigraph Y is connected: otherwise some component of Y is joined to C by at most two edges, and these form a cut. So Y is a 4-pole with the ports $1, 2, 3, 4$. Let k be its rank. Then $r = k + 4$.

Let e be a bridge of Y, and let A and B be the vertex sets of the two components of $Y - e$. In $G[A]$ every vertex has degree at least two, except the end of e, which has degree at least one. A forest with an edge has two vertices of degree one, so $G[A]$ contains a cycle. The same holds for $G[B]$. Each of the four edges between C and Y is in exactly one of the cuts δA and δB, and e is in both. So $|\delta A| + |\delta B| = 6$, and one of them is a cyclic cut of size at most three. Hence Y has no bridge.

The count. We use the notation above for Y, with $h = k$ and $z = 2^{-k} c(Y)$. A cycle of G that is neither C nor a cycle of Y uses two or four of the edges between C and Y. With two, at c_i and c_j, it consists of a path of Y between the ports i and j and of one of the two paths of C between c_i and c_j. With four, it consists of a pair of disjoint paths of Y and of two disjoint edges of C. The edges c_1c_2 and c_3c_4 close the pair to one cycle if and only if its pairing is not $12|34$. The edges c_1c_4 and c_2c_3 do this if and only if its pairing is not $14|23$. Hence

$$\frac{c(G)}{2^k} \;=\; 2^{-k} + z + \bigl(S + F - u_{12|34}\bigr) + \bigl(S + F - u_{14|23}\bigr). \tag{3.4}$$

Let r be so large that $k \geq h_1$ and $2^{-k} \leq \varepsilon_0$. Suppose that $c(G) \geq 2^{r-1} = 8 \cdot 2^k$. We show that G is a ring. By Lemma 3.1 for the pairing $14|23$, the term z and the second bracket of (3.4) are together at most $4 + \varepsilon_0$. So $S + F - u_{12|34} \geq 4 - 2\varepsilon_0$. By Lemma 3.1 for the pairing $12|34$, this gives $z \leq 3\varepsilon_0$. Also $S + F \geq 4 - 2\varepsilon_0$, so Lemma 3.2 applies to Y. It gives two cases.

Case 1: three pairs with a common port are full. By (3.1), the other three numbers a_{ij} are at most κ, so $S \leq 3 + 3\kappa$. By [M, Lemma 6.7], every number u_α is at most $\frac{1}{2}$. The sum of the two brackets in (3.4) is $2S + F + u_{13|24}$, and this is at most $6 + 6\kappa + \frac{3}{2}$. So $c(G)\, 2^{-k} \leq \frac{15}{2} + 10\varepsilon_0 < 8$. This case does not occur.

Case 2: the four pairs between two disjoint pairs of ports are full. The pole Y has no bridge. By [M, Corollary 5.11] it is a chain of tiles; one of the two pairs of ports is the left pair of its first tile, and the other is the right pair of its last tile. If the pairs are $1, 3$ and $2, 4$, then C is a square with the pairs c_1, c_3 and c_2, c_4. In the other case let the pairs be $1, 2$ and $3, 4$, by symmetry. Then the edges c_1c_2 and c_3c_4 are single rungs, and the edges c_2c_3 and c_1c_4 are the interface between them. In both cases the edges between C and Y are two interfaces. So G is a ring. ∎

IDEA | a comparison with the proof of [M, Theorem 6.8]

In [M] the cut has two sides X and Y of ranks $h \le k$. [M, Lemma 6.4] counts the cycles across the cut by the states of both sides and gives the estimate $\mathcal{B} \le S(Y) + F(Y) - \min_\alpha u_\alpha(Y)$. Then both cases of the proof first show that the pairs that are full in Y are full in X, with the gap theorem for X, and then use [M, Lemma 6.7] or [M, Corollary 5.11] for both sides. Here the side X is the cycle C. Its states are known: every $a_{ij}(C)$ is 1, and $u_\alpha(C)$ is $\frac{1}{2}, 0, \frac{1}{2}$. So the count is the equation (3.4), there is no step for the second side, and the cap is used for the two pairings of (3.4).

4 Three cases

Let G be a connected cubic multigraph of rank $r \ge 2$ with a loop at a vertex v. The loop uses two of the three edge ends at v. So v has one other edge e, the set $\delta\{v\}$ consists of e, and it is a cyclic cut of size one: one side contains the loop, and the other side has rank $r-1$. The case $d = 1$ in the proof of [M, Proposition 6.3] uses no property of the sides: if δX is a cut of size one, then every cycle lies in one side, the ranks b_1, b_2 of the sides are positive with $b_1 + b_2 = r$, and $c(G) \le (2^{b_1} - 1) + (2^{b_2} - 1) \le 2^{r-1}$. So [M, Proposition 6.3] holds for multigraphs with loops, with one more sentence in its proof: if the least size d of a cyclic cut is two or three, then G has no loop, and the sides are poles. Case 1 of [M, Section 9] is then a part of Case 2.

With [N2, Theorem 5.1] and Theorem 3.3, the proof of the main theorem reads as follows. Let G be a connected cubic multigraph of large rank r with $\psi(r)$ cycles.

1. *G has a cyclic cut of size at most three.* Then $c(G) \le 2^{r-1} + 5K_0^2\, 2^{r/2}$ by [M, Proposition 6.3].
2. *G has no such cut, and it has a cycle of length four.* Then G is a ring or $c(G) < 2^{r-1}$, by Theorem 3.3.
3. *G has no such cut and no cycle of length four.* A cycle Z with at most three edges would give the cyclic cut $\delta V(Z)$ of size at most three. So G has girth at least five. Choose a set P of eight vertices such that $G[P]$ is connected. Then $q(G,P) < \frac{2}{5}$ by [N2, Theorem 5.1]. The graph has no bridge, and Lemma 5.1 below gives $c(G) < \frac{2}{5}\, 2^r + 2^{r-5} < 2^{r-1}$.

5 The count with a marked set

Recall that $q(G,P)$ is the probability that the edges of a random even subgraph without an end in P form a forest.

Lemma 5.1 (Cycles and a marked set)**.** Let G be a connected cubic multigraph of rank r without a bridge. Let P be a set of p vertices, not all vertices of G, such that $G[P]$ is connected. Then

$$c(G) \;\le\; q(G,P)\, 2^r + 2^{\,r-1-p/2}.$$

Proof. This is the count in the last case of [M], for p in the place of 8. Let Z be a cycle of G that meets P. Its edges in $G - P$ form a proper subset of Z, hence a forest. So at most $q(G,P)\,2^r$ cycles meet P. The other cycles are cycles of $G - P$, and there are at most $2^{\beta(G-P)}$ of them. Let a be the number of edges of $G[P]$. Then $3p - 2a$ edges leave P, and $G - P$ arises from G by the deletion of p vertices and $3p - a$ edges. So $\beta(G-P) = r - 1 - 2p + a + \mathrm{comp}(G-P)$. The graph G has no bridge, so every component of $G - P$ contains the ends of at least two of the edges that leave P. Hence $\mathrm{comp}(G-P) \le (3p-2a)/2$ and $\beta(G-P) \le r - 1 - p/2$. ∎

Corollary 5.2 (Leading term; [M, Corollary 3.7]). There is r_1 such that every graph of rank $r \ge r_1$ has at most $\left(\frac{1}{2} + 21/\log^{(3)} r\right)2^r$ cycles.

Proof. Let r be large, and let G be a connected cubic multigraph of rank r with $\psi(r)$ cycles ([M, Lemma 3.6]). Suppose first that G has a bridge e, and let b_1 and b_2 be the ranks of the two components of $G-e$. A component with n vertices has $(3n-1)/2 \ge n$ edges, so $b_1, b_2 \ge 1$, and $b_1 + b_2 = r$. Every cycle lies in one component. So $c(G) \le (2^{b_1} - 1) + (2^{b_2} - 1) \le 2^{r-1}$.

In the other case choose a set P of $\lceil 4 \log r \rceil$ vertices such that $G[P]$ is connected. Then $|P| \le 2r/\log^2 r$. By [M, Theorem 3.1] and Lemma 5.1,

$$c(G) \;\le\; \left(\frac{1}{2} + \frac{20}{\log^{(3)} r}\right)2^r + 2^{\,r-1-2\log r} \;\le\; \left(\frac{1}{2} + \frac{21}{\log^{(3)} r}\right)2^r.$$ ∎

The proof in [M] takes a shortest cycle W, shows that it has at most $2\log(2r) + 2$ vertices, counts the cycles through one edge e of W with the map $C \mapsto C + W$, and solves the recursion $\psi(r) \le \psi(r-1) + (\frac{1}{2} + \delta_r)2^{r-1}$. The proof above has no shortest cycle and no recursion. In typeset lines the two forms are equal: Section 3 has seven lines more, and the proof of the main theorem has nine lines less. The gain is that one count serves two places.

6 Accounting

6.1 Measured copies

The numbers below are for built copies of the paper. A typeset line is a line of the text that `pdftotext` gives for the body of the paper. "Lean" and "clean" are the two forms of [N2].

	Now	[N2] lean	Lean, 2–4	Lean, 1–4	Clean, 1–4
Pages	42	42	42	41	41
Typeset lines	1624	1617	1607	1587	1610
Numbered statements	40	40	40	39	40
Cases in the proof of the main theorem	5	4	3	3	3
Section 3 (forest estimate), lines	265	267	272	272	
Section 5 (poles), lines	339	301	300	276	
Section 6 (rings and cuts), lines	275	276	261	258	
Proof of the main theorem, lines	30	31	20	20	

"Lean, 1–4" is the lean form of [N2] with the four reductions. The file `trial-fewer-cases.pdf` is this copy. It has one new statement (Lemma 5.1) and does not have [M, Lemma 5.9] and [M, Lemma 6.4]. "Lean, 2–4" is the copy without the first reduction (the mode `keepgap` of the script): the gap theorem keeps its proof by induction and the constant $\frac{1}{16}$, and Section 6 keeps its numbers. So the first reduction gives 20 of the 30 lines, and the other three change the cases and the statements.

6.2 Mathematical steps that the copy does not have

- The lemma on a third port with its four positions of the port, and the count of paths from an inner vertex of a capped ladder.
- The induction on the rank in the gap theorem, with the deletion of a port.
- The count for two 4-poles, and the estimate of $\mathcal{B}$ by the states of one side.
- In the theorem on a cut of size four: the proof that both sides are 4-poles without bridge (now one side), and in each of the two cases the step "the pairs are full in X as well".
- The case of a loop.
- The bound for the length of a shortest cycle, the map $C \mapsto C + W$, and the recursion in the proof of the leading term.

6.3 What is new in the copy

- Item (iv) of [M, Proposition 5.3] (Lemma 2.1), with one sentence in Step 2 of its proof.
- The equivalence of three statements in the lemma on ladders, in the place of one implication. The proofs are those of [M].
- The count (3.4) for a cycle of length four, and a figure with the two caps.
- Lemma 5.1. Its proof was a part of the proof of the main theorem.
- The symbol ε_0 in the place of the numbers of Section 6.

6.4 The price

- *Dependence.* In [M] the gap theorem and [M, Lemmas 5.8 and 5.9] are elementary: they do not use the forest estimate. The new proof of the gap theorem uses [M, Proposition 5.3], hence [M, Lemma 4.1] and [M, Theorem 3.1]. The main theorem uses these in every case, so no result of the paper becomes weaker. But the gap theorem alone is no longer an elementary statement with the constant $\frac{1}{16}$.
- *Constants.* $\gamma_0 = 2^{-20\Theta}$ with $\Theta = \Theta_{1/8}(1) \geq 2^{102}$, and $K_0 = K_{\varepsilon_0}$ with $\varepsilon_0 = \gamma_0/20$. The theorem is for large r without a value of R, as before.
- *The formal proof.* The formal proof has the gap with the constant $\frac{1}{16}$ by induction, the count for two sides, a cut of size four in the place of a cycle of length four, and the edge recursion. The trial copy says this in its Section 1.4: six places with another method, in the place of four.
- *Reduction 2 needs [N2].* Without the theorem for girth five, a graph with a cyclic cut of size four and no cycle of length four has no case.

7 What I tried without success

I record these attempts so that nobody repeats them without a new idea.

1. *No proposition for small cuts.* With the theorem for girth five, a graph of large rank without a bridge and of girth at least five has fewer than 2^{r-1} cycles. So one can order the cases by short cycles: a cycle of length two, three or four. This does not remove [M, Proposition 6.3]. Two capped ladders, joined at their caps by two edges, form a graph with a cut of size two and $2^{r-1} + c(T_1) + c(T_2)$ cycles. So a bound of the form $2^{r-1} + O(2^{r/2})$ is necessary for cuts with two large sides, and it needs [M, Theorem 5.4] with the exponent $b/2$. A graph with a cycle of length four can also have such a cut, so Theorem 3.3 needs the hypothesis on cuts.
2. *An induction on the rank.* Close a 2-pole T of rank b by an edge between its ports. This gives $c(T)+P(T) \le \psi(b+1)$, and for a cut of size two it would give $c(G) \le 2^{r-1}+(\psi(b_1+1)-2^{b_1})+(\psi(b_2+1)-2^{b_2})$. With $\psi = H$ on the right side this is less than $H(r)$. The equation (3.4) also has this form: the count in the proof of Lemma 3.1 is exact, so $c(G) + c(Y) = 1 + c(\hat{Y}_{12|34}) + c(\hat{Y}_{14|23})$, and the two capped graphs have rank $r-1$. But the main theorem is false for small rank (the example of rank 27 in [M]), so an induction has no start. The text of [M] removed an earlier induction for the same reason.
3. *Rigidity with fewer tools.* The proof of Lemma 3.2 uses (3.1), (3.2), (3.3), and Theorem 3.3 also uses the cap and the lemma on a full star. Each of them is necessary for this arithmetic. The state $a_{13} = a_{14} = a_{23} = 1$, $a_{24} = \frac{1}{2}$, $u_{12|34} = \frac{1}{2}$ (all other numbers 0) has $S + F = 4$ and satisfies (3.1), (3.2) and the three cap inequalities; only (3.3) excludes it. The state $a_{13} = a_{14} = a_{23} = 1$, $u_{14|23} = 1$ satisfies (3.1), (3.3) and the caps; only (3.2) excludes it. The star $a_{12} = a_{13} = a_{14} = 1$, $u_{13|24} = 1$ satisfies all of them and gives the value 8 in (3.4); only the lemma on a full star excludes it. Test E of the script verifies these three statements.
4. *A gap theorem with a good constant and without induction.* If $P_{st}(N) > (1-\gamma)2^b$, then every cycle of J has more than $\log(1/\gamma)$ vertices. I found no elementary reason why J then has no cycle; the argument through the layers needs the forest estimate and gives the small constant.
5. *A polynomial bound for pieces at girth five without induction.* A thin piece has many disjoint cycles of length $O(\log k)$ far from its boundary. If the joins can avoid the edges around many of them, [M, Lemma 2.4] gives the bound. If not, each of these cycles separates the set T, and the piece is a chain of blocks with small boundaries; the bound for a block is the statement for a smaller k. So the induction of [N2, Theorem 4.2] with a base for bounded k is the natural form. The base needs a statement on rigid sets, because a block of bounded size gives only a factor $1 - 2^{-\mathrm{rank}}$.
6. *The base without rigid sets.* For a set T with forest proportion more than $\frac{1}{2} + \varepsilon$, the marker argument and [M, Lemma 2.4] show that the graph outside the marker has bounded rank, as in [M, Proposition 5.3]. For two boundary vertices this gives the ladder. For more boundary vertices the count gives only that about half of the vertices are outside the marker, and I found no cycle of length four from this.
7. *One lemma for the marker.* Step 1 of the proof of [M, Proposition 5.3] and the step "a marker" in the lemma on pieces are the same argument. One lemma can serve both. This is a change of the presentation only, and it saves no step, so the trial copy does not have it.

8. *The last section.* I found nothing new for the chain, the heavy families and the packing bound after [N1] and [N2]. The threshold $\frac{2}{5}$ can be any number below $\frac{1}{2}$ if the marked set is larger (Lemma 5.1), but the proof needs two pieces for every such threshold, so this does not simplify it.

8 Tests

The script `checks/gap_and_square_check.py` has five tests. Exit code 0 means that all of them passed. With the arguments `1500 1` it prints:

Test	Content and result
A	1500 random bridgeless 2-poles of rank 2 to 6, 206 with a full pair. For every shortest path A: the graph J is a forest if and only if the pair is full (454 and 1578 paths). If J has a cycle Z, then $P_{st} \le 2^b - 2^{b-\lvert Z\rvert+k-1}$.
B	50 cubic graphs with 10 to 14 vertices and a cycle of length four. The count behind equation (3.4) and the identity $c(G)+c(Y) = 1 + c(\hat{Y}_{12\lvert 34}) + c(\hat{Y}_{14\lvert 23})$ agree with a direct count of the cycles.
C	50 cubic graphs without a bridge and a connected set P of 2 to 6 vertices: $q(G,P)$ by enumeration of all even subgraphs; Lemma 5.1 and the bound for $\beta(G-P)$ hold.
D	The arithmetic of Lemma 3.2: 181 local searches over the states that satisfy (3.1)–(3.3) and do not have the conclusion of the lemma, for $\eta \in \{\frac{1}{80}, \frac{1}{400}, \frac{1}{5000}\}$. The largest value of $S+F$ was $4-2\eta-0.507\eta$, below $4-2\eta$.
E	The three states of Section 7, item 3.

The tests A, B, C are complete counts on small graphs. They test the statements that do not contain large constants. Theorem 2.3 and Theorem 3.3 are statements with large constants, and no computation tests them.

9 Limits

- The proofs are on paper. There is no formal proof of the new statements, and no second reader.
- The new gap theorem is weaker than [M, Theorem 5.10] as a statement: its constant is smaller. The paper uses only that the constant is positive and fixed.
- The four reductions change Sections 3, 5, 6 and 9. They do not change the last section, which is the longest one.
- I did not build a copy of the present text with the reductions 1, 3 and 4 alone. For that copy the proof of [M, Theorem 6.8] needs the numbers of Section 2.4; the changes are of the same kind as in Lemma 3.2.

10 Files

File	Content
`research/fewer-cases.tex, .pdf`	this note
`research/apply_fewer_cases.py`	writes the four reductions into a copy that `apply_girth_five.py` has changed; with the argument `keepgap` it writes the reductions 2, 3 and 4 only; every replacement must match exactly once
`research/trial-fewer-cases.pdf`	the copy: lean form of [N2] and the four reductions, 41 pages
`checks/gap_and_square_check.py`	the five tests of Section 8

To build the copy, copy the folders `exact-maximum` and `style` to a new folder. On the copy of `exact-maximum` run the two scripts in this order:

```
python3 apply_girth_five.py <copy> lean
  python3 apply_fewer_cases.py <copy>
```

Then build `exact-maximum.tex` with `tectonic`.

References

[M] KNT, Astra and Fable 5.2, *The maximum number of cycles in a graph of large rank*, draft of 12 November 2026, 42 pages, `exact-maximum/exact-maximum.pdf`.

[N1] Fable 5.2, *The large-girth theorem with one family of short cycles*, research note of 12 November 2026, `research/one-scale.pdf`.

[N2] Fable 5.2, *Pieces at girth five: the case of bounded girth is not necessary*, research note of 12 November 2026, `research/girth-five.pdf`.